Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/26/2024
PI Name:	Dmochowski, Jacek Ph.D.		
Project Title:	Boosting Brain Metabolism in Spaceflight with Transcranial Photobiomodulation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031-9129	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	08/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/30/2021 per E. Urquieta/TRISH (Ed., 6/9/21) NOTE: End date changed to 06/30/2021 per TRISH (Ed., 7/6/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liu, Hanli Ph.D. (The University of Texas System)		
Grant/Contract No.:	NNX16AO69A-T0409		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this proposal is to develop a new technique for enhancing brain activity by applying safe levels of laser light to the forehead. It is known that near-infrared light is absorbed by an enzyme in the mitochondria and subsequently leads to greater energy availability in the stimulated cells. When stimulating the brain, this additional energy can be used by brain cells to carry out the necessary functions required to maintain high cognitive performance (for example, sustained attention, accurate memory recall). We will test the proposed technology by scanning participants' brain activity while they receive laser stimulation to the forehead and at the same time perform a challenging perceptual task. We will also perform experiments where Light Emitting Diodes (LEDs) are used to stimulate participants' brains while they are trained on a working memory task and simultaneously have their brain activity measured. Our expectation is that we will observe increased energy metabolism in the frontal part of the brain, as well as an improvement in the performance of both perceptual and memory tasks. We expect that this research will lead to a new, non-pharmacological technique for boosting the performance of the brain by delivering safe levels of near-infrared light to the brain.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technology being developed has potential utility in spaceflight, investigative treatments for depression and other vascular related disorders, as well as cognitive enhancement. We expect that the publications stemming from this project will become highly cited and spur the development of transcranial photobiomodulation (tPBM) in basic and clinical neuroscience.
Task Progress:	We performed the first in-human experiment combining P31 MRS with tPBM. We found a significant decrease in the PCr/ATP ratio in the active stimulation group, indicating that the concentration of ATP increased during active stimulation (relative to sham). These results are significant because they are consistent with the dominant hypothesis in photobiomodulation: that photons of near-infrared light are absorbed by the mitochondrial enzyme CCO, accelerating the production of ATP and hence increasing the energy availability of the cell. As such, these findings represent direct evidence for the ability of near-infrared light to modulate cerebral bioenergetics in the human brain. Impact statement: the findings of this study pave the way for comprehensive investigations of cognitive performance using tPBM in more ecologically valid spaceflight environments. The ease-of-use of LED, along with advantages of being cheap, having low dose, and being harmless makes it a preferable choice of conducting tPBM. This is the first study, to our knowledge, that demonstrates the effects of LED stimulation on the prolonged improvement of attention performance and acute boost of electrophysiology functions on healthy subjects. We conducted 64-channel EEG measurements from 22 healthy human subjects while they performed the PVT task. Improvement in vigilance/attention was tracked (reaction time) over four weeks. We found a significant increase in reaction time of tPBM treated subjects over time. In addition, we found LEDs have significant effects on global EEG power at the temporal lobe and motor region in the delta band, and left temporal, parieto-occipital lobes in the alpha band in the human brain. Impact statement: the findings of this study indicate that LEDs may be used in place of lasers for tPBM, thus alleviating the technological burden imposed by laser-based systems.
Bibliography Type:	Description: (Last Updated: 03/13/2025)